THE BEST INEQUALITY FOR NONCOMMUTATIVE CONDITIONALLY SYMMETRIC MARTINGALES

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Abstract. We establish square-function inequalities with optimal growth order for noncommutative conditionally symmetric martingales. For general non-self-adjoint martingales, conditional symmetry is formulated through self-adjoint dilation, which coincides with the usual conditional symmetry in the self-adjoint case. We prove that, for every $p \geq 4$, there exists an absolute constant $C > 0$ such that

$$\|x_N\|_p \leq C\sqrt{p}\,\max\{\|S_c(x)\|_p, \|S_r(x)\|_p\}.$$

We further show that the best constants for the self-adjoint and matrix-conditionally symmetric classes coincide, and that the order $\sqrt{p}$ is optimal as $p \to \infty$. Our approach combines soft truncations, Cuculescu projections, and localized exponential estimates. Applications to several noncommutative martingale models are also presented.

1. Introduction

The Burkholder–Gundy inequalities [4] are basic results in martingale theory. They compare the L_p norm of a martingale with the L_p norm of its square function. For a scalar martingale $f = (f_n)_{n=0}^N$ with $f_0 = 0$, they give

$$c_p\|f_N\|_p \leq \|S(f)\|_p \leq C_p\|f_N\|_p, \qquad 1 < p < \infty,$$

where $S(f) = (\sum_{n=1}^N |d_n f|^2)^{1/2}$. An important question is how the best constants depend on p. This was studied in several works of Burkholder [2, 3]. $O(\alpha_p) =?, p \to 1, p \to \infty$ Better bounds may hold when the martingale differences have more structure. In particular, dyadic martingales, which are closely related to the Haar system and dyadic harmonic analysis, are conditionally symmetric: The conditional distributions of $d_n f$ and $-d_n f$ given $\mathcal{F}_{n-1}$ coincide. Wang [23] proved sharp square-function inequalities for Hilbert-space-valued conditionally symmetric martingales and obtained results for Haar functions. Further sharp weak-type estimates were studied by Osekowski [14]. These results show that symmetry can lead to better constants, and they motivate the same question for noncommutative martingales.

For a martingale $x = (x_n)_{n=0}^N$ in a noncommutative $L_p(\mathcal{M})$ space, the differences and their adjoints need not commute. We therefore use the column and row square functions

$$S_c(x) = \Big(\sum_{n=1}^N (d_n x)^* d_n x\Big)^{1/2}, \qquad S_r(x) = \Big(\sum_{n=1}^N d_n x (d_n x)^*\Big)^{1/2},$$

where $d_n x = x_n - x_{n-1}$. This distinction also appears in the noncommutative Khintchine inequalities of Lust-Piquard and Lust-Piquard–Pisier [12, 13]. Pisier and Xu [15] proved the noncommutative Burkholder–Gundy inequalities and used them to extend the theory of the Itô–Clifford integral from L_2 to L_p. Further results include the noncommutative Burkholder–Rosenthal inequalities of Junge and Xu [10] and weak-type

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square-function estimates of Randrianantoanina [18]. Junge and Xu [11] also determined the optimal growth of the constants in several noncommutative martingale inequalities. In particular, the best constant β_p in

$$\|x_N\|_p \le \beta_p \max\{\|S_c(x)\|_p, \|S_r(x)\|_p\}$$

for general noncommutative martingales satisfies $\beta_p \asymp p$ as $p \to \infty$. The earlier results apply to Clifford martingales, but do not give the $\sqrt{p}$ order for this class. This leaves the question of whether the smaller order known for classical symmetric martingales can also hold in the noncommutative case.

In the classical setting, sharp martingale bounds can be studied using Bellman functions [3, 23], comparisons with tangent or decoupled sequences [7], and exponential tail estimates followed by good-λ arguments [4, 8]. These ideas do not apply to our problem without changes. For instance, stopping a noncommutative martingale uses projections, and compression by a projection from the past need not preserve conditional symmetry. Noncommutative Azuma-type estimates and good-λ methods are available [9, 21], but they do not directly give the local bound we need. Our proof uses a different combination of these ideas. We first make odd soft truncations of the differences, so they remain martingale differences. We then use adapted Cuculescu projections [5, 17] to separate changes in the truncation from changes in the projections. We control one part in operator norm and use a local L_2 bound and an exponential estimate for the other. Finally, we sum over the truncation scales. In this way, we obtain the $\sqrt{p}$ bound without using a good-λ inequality or assuming that symmetry is preserved by compression.

Clifford martingales were one starting point for this problem, but the proof does not depend on the CAR relations. We work with the larger class of noncommutative conditionally symmetric martingales. For non-self-adjoint martingales, we use a self-adjoint matrix dilation. This equals the usual condition for self-adjoint martingales and also covers non-self-adjoint examples. We give equivalent characterizations of conditional symmetry and a sufficient criterion based on trace-preserving $*$-automorphisms of $\mathcal{M}_n$ that fix $\mathcal{M}_{n-1}$ pointwise. This criterion is not required in our main theorem. It applies to Clifford, operator-valued dyadic, and nest-filtration martingales [15, 16], showing that our result covers several noncommutative settings.

We now state the main result. Let $(\mathcal{M}, \tau)$ be a von Neumann algebra with a normal faithful semifinite trace, and let $(\mathcal{M}_n)_{n=0}^N$ be an increasing filtration with normal trace-preserving conditional expectations $\mathcal{E}_n$. We assume that the restricted traces are semifinite. For a self-adjoint martingale x, write

$$S(x) = \left(\sum_{n=1}^{N} (d_n x)^2\right)^{1/2}.$$

In this case, conditional symmetry means

$$\mathcal{E}_{n-1}\left(f(d_n x)\right) = \mathcal{E}_{n-1}\left(f(-d_n x)\right), \qquad f \in C_b(\mathbb{R}), \quad 1 \le n \le N. \tag{1.1}$$

For a general martingale, define

$$\hat{x}_n = \begin{pmatrix} 0 & x_n \\ x_n^* & 0 \end{pmatrix}, \qquad \widehat{\mathcal{E}}_n = \mathrm{id}_{M_2} \otimes \mathcal{E}_n.$$

We call x (matrix) conditionally symmetric if

$$\widehat{\mathcal{E}}_{n-1}\left(f(d_n \hat{x})\right) = \widehat{\mathcal{E}}_{n-1}\left(f(-d_n \hat{x})\right), \qquad f \in C_b(\mathbb{R}), \quad 1 \le n \le N. \tag{1.2}$$

This is equivalent to (1.1) when x is self-adjoint. For general x, it does not require equality of the full conditional $*$-distributions of $d_n x$ and $-d_n x$. We discuss this definition in Section 2.2.

Theorem 1.1. *Let $(\mathcal{M},\tau)$ be a s.n.f. von Neumann algebra, and let $x=(x_n)_{n=0}^N$ be a conditionally symmetric $L_p(\mathcal{M})$-martingale with $x_0=0$. There exists an absolute constant $C>0$ such that, for every $4\le p<\infty$,*

$$\|x_N\|_p \le C\sqrt{p}\,\max\{\|S_c(x)\|_p, \|S_r(x)\|_p\}. \tag{1.3}$$

In particular, if x is self-adjoint and satisfies (1.1), then

$$\|x_N\|_p \le C\sqrt{p}\,\|S(x)\|_p. \tag{1.4}$$

The constant is independent of p, N, and $\mathcal{M}$.

We also show that the order $\sqrt{p}$ is best possible. Let C_p^{best} be the least constant in

$$\|x_N\|_p \le C_p^{\mathrm{best}}\max\{\|S_c(x)\|_p, \|S_r(x)\|_p\}$$

for the martingales in Theorem 1.1, and let $C_{p,\mathrm{sa}}^{\mathrm{best}}$ be the least constant in the corresponding self-adjoint estimate.

Corollary 1.2. *For every $4\le p<\infty$,*

$$C_p^{\mathrm{best}} = C_{p,\mathrm{sa}}^{\mathrm{best}}, \qquad c\sqrt{p}\le C_p^{\mathrm{best}}\le C\sqrt{p}, \tag{1.5}$$

where $c,C>0$ are absolute constants.

The equality shows that self-adjoint dilation does not change the best constant. The bounds give the optimal order as $p\to\infty$; they do not identify the exact best constant. For $1<p<4$, the corresponding estimate follows from the noncommutative Burkholder–Gundy inequalities [11, 15, 18], with a possibly larger absolute constant. The paper is organized as follows. Section 2 gives the definitions and tools used below. Section 3 contains the proof of the main theorem. Section 4 gives four classes of examples.

2. Preliminaries

2.1. **Measurable operators and noncommutative L_p spaces.** Let $\mathcal{M}$ be a von Neumann algebra with a normal faithful semifinite trace τ. Let $L_0(\mathcal{M})$ denote the algebra of all τ-measurable operators. For $1\le p\le\infty$, let $L_p(\mathcal{M},\tau)$ (simply $L_p(\mathcal{M})$) be the associated noncommutative Lebesgue space. As usual, $L_\infty(\mathcal{M})$ is just $\mathcal{M}$ with the usual operator norm. Also recall that

$$L_p(\mathcal{M}) = \{x\in L_0(\mathcal{M}) : \|x\|_{L_p(\mathcal{M})}<\infty\}$$

with

$$\|x\|_{L_p(\mathcal{M})} = [\tau(|x|^p)]^{1/p}, \qquad 1\le p<\infty,$$

where $|x|=(x^*x)^{1/2}$ is the modulus of x.

Suppose that a is a self-adjoint τ-measurable operator and let $a=\int_{-\infty}^{\infty}\lambda de_\lambda$ stand for its spectral decomposition. For any Borel subset B of $\mathbb{R}$, the spectral projection of a corresponding to the set B is defined by $\chi_B(a)=\int_{-\infty}^{\infty}\chi_B(\lambda)de_\lambda$. We also use polar decomposition. For each $x\in L_0(\mathcal{M})$, let $x=u|x|$ be its polar decomposition. By the spectral calculus (see [22, (23)]),for every bounded Borel function $g:[0,\infty)\to\mathbb{C}$, we have

$$ug(|x|) = g(|x^*|)u, \qquad u^*g(|x^*|) = g(|x|)u^*. \tag{2.1}$$

Indeed, writing $q=uu^*$, the polar decomposition gives

$$g(|x^*|) = ug(|x|)u^* + g(0)(1-q),$$

which implies both identities.

We shall repeatedly use the self-adjoint operator inequality

$$(x-y)^2 \le \frac{5}{4}x^2 + 5y^2. \tag{2.2}$$

Indeed, the difference between the right-hand side and the left-hand side is $(x/2+2y)^2$, which is positive.

2.2. **Noncommutative conditionally symmetric martingales.** Equip

$$\widehat{\mathcal{M}} = M_2(\mathbb{C})\,\overline{\otimes}\,\mathcal{M}, \qquad \widehat{\mathcal{M}_n} = M_2(\mathbb{C})\,\overline{\otimes}\,\mathcal{M}_n$$

with the trace $\hat{\tau} = \mathrm{tr}_2 \otimes \tau$, where tr_2 is the normalized matrix trace, and with the conditional expectations $\widehat{\mathcal{E}_n}$ introduced above.

Definition 2.1. A finite L_p-martingale x is called *conditionally symmetric* if its self-adjoint dilation $\hat{x}$ satisfies (1.2). Here $C_b(\mathbb{R})$ denotes the bounded continuous complex-valued functions on $\mathbb{R}$.

Lemma 2.2. *Let $(\mathcal{M},\tau)$ be a s.n.f. von Neumann algebra, and let $x \in L_0(\mathcal{M})$ have polar decomposition $x = u|x|$. For each bounded Borel function f, we have*

$$f\begin{pmatrix} 0 & x \\ x^* & 0 \end{pmatrix} = \begin{pmatrix} f_1(|x^*|) & uf_2(|x|) \\ f_2(|x|)u^* & f_1(|x|) \end{pmatrix}$$

where $f_1(t) = \frac{1}{2}\left(f(t)+f(-t)\right)$ and $f_2(t) = \frac{1}{2}\left(f(t)-f(-t)\right)$.

Proof. Let $x = u|x|$ denote the polar decomposition. Let $A(x)$ denote $\begin{pmatrix} 0 & x \\ x^* & 0 \end{pmatrix}$. Note that

$$A(x)A(x)^* = \begin{pmatrix} |x^*|^2 & 0 \\ 0 & |x|^2 \end{pmatrix} = \begin{pmatrix} |x^*| & 0 \\ 0 & |x| \end{pmatrix}^2.$$

So $|A(x)| = \begin{pmatrix} |x^*| & 0 \\ 0 & |x| \end{pmatrix}$. Let $A(x) = \mathrm{sgn}(A)|A(x)|$ denote the polar decomposition of $A(x)$. Since

$$\begin{pmatrix} 0 & u \\ u^* & 0 \end{pmatrix}|A(x)| = A(x)$$

and

$$\begin{pmatrix} 0 & u \\ u^* & 0 \end{pmatrix}^2 = \begin{pmatrix} s(|x^*|) & 0 \\ 0 & s(|x|) \end{pmatrix} = s(|A|),$$

we have $\mathrm{sgn}(A) = \begin{pmatrix} 0 & u \\ u^* & 0 \end{pmatrix}$. Here, we use the uniqueness of polar decomposition. Since f_1 is even, we have $f_1(t) = f_1(|t|)$ and thus,

$$f_1(A(x)) = f_1(|A(x)|) = \begin{pmatrix} f_1(|x^*|) & 0 \\ 0 & f_1(|x|) \end{pmatrix}.$$

For f_2, we have $f_2(t) = \mathrm{sgn}(t)f_2(|t|)$. Hence,

$$f_2(A(x)) = \begin{pmatrix} 0 & u \\ u^* & 0 \end{pmatrix}\begin{pmatrix} f_2(|x^*|) & 0 \\ 0 & f_2(|x|) \end{pmatrix} = \begin{pmatrix} 0 & uf_2(|x|) \\ u^*f_2(|x^*|) & 0 \end{pmatrix}.$$

Combining these identities and using (2.1), we get

$$f(A(x)) = \begin{pmatrix} f_1(|x^*|) & uf_2(|x|) \\ f_2(|x|)u^* & f_1(|x|) \end{pmatrix}.$$

□

The next proposition shows both that this extends the original self-adjoint condition and that it is a condition on each difference $d_n x$ itself.

Proposition 2.3. *For a martingale $x = \{x_n\}_{n=0}^N$, the following assertions hold.*

(i) *If $x = \{x_n\}_{n=0}^N$ is self-adjoint, then (1.2) is equivalent to (1.1).*

(ii) *Let $u_n|d_n x|$ denote the polar decomposition of $d_n x$. (1.2) is equivalent to*

$$\mathcal{E}_{n-1}\left(u_n \varphi(|d_n x|)\right) = 0 \quad \text{for every } \varphi \in C_b([0,\infty)) \text{ with } \varphi(0) = 0, \tag{2.3}$$

for each n.

(iii) *If the martingale differences $\{d_n x\}_{n=1}^N$ are bounded, conditional symmetry is also equivalent to*

$$\mathcal{E}_{n-1}\left(d_n x\left((d_n x)^* d_n x\right)^k\right) = 0, \qquad k = 0, 1, 2, \ldots, \quad 1 \le n \le N. \tag{2.4}$$

Proof. (i) Consider the scalar unitary matrix $U = 2^{-1/2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$. For a self-adjoint $x = \{x_n\}_{n=0}^N$, we have

$$U\widehat{\mathcal{E}}_{n-1}\left(f(d_n \hat{x})\right) U^* = \widehat{\mathcal{E}}_{n-1}\left(f(U d_n \hat{x} U^*)\right).$$

Here, we use unitary covariance of functional calculus and the bimodule property of conditional expectation. Note that

$$U\begin{pmatrix} 0 & d_n x \\ d_n x & 0 \end{pmatrix} U^* = \begin{pmatrix} d_n x & 0 \\ 0 & -d_n x \end{pmatrix},$$

and hence,

$$U\widehat{\mathcal{E}}_{n-1}\left(f(d_n \hat{x})\right) U^* = \begin{pmatrix} \mathcal{E}_{n-1}\left(f\left(d_n x\right)\right) & 0 \\ 0 & \mathcal{E}_{n-1}\left(f\left(-d_n x\right)\right) \end{pmatrix}.$$

Similarly, we have

$$U\widehat{\mathcal{E}}_{n-1}\left(f(-d_n \hat{x})\right) U^* = \begin{pmatrix} \mathcal{E}_{n-1}\left(f\left(-d_n x\right)\right) & 0 \\ 0 & \mathcal{E}_{n-1}\left(f\left(d_n x\right)\right) \end{pmatrix}.$$

Thus (1.2) is equivalent to (1.1).

(ii) Let $\tilde{\varphi}$ denote the odd extension on $\mathbb{R}$ of φ. According to Lemma 2.2, we can directly get

$$\widehat{\mathcal{E}}_{n-1}\left(\tilde{\varphi}\left(d_n \hat{x}\right)\right) = \begin{pmatrix} 0 & \mathcal{E}_{n-1}\left(u_n \tilde{\varphi}(|d_n x|)\right) \\ \mathcal{E}_{n-1}\left(\tilde{\varphi}(|d_n x|) u_n^*\right) & 0 \end{pmatrix}$$

and

$$\widehat{\mathcal{E}}_{n-1}\left(\tilde{\varphi}\left(-d_n \hat{x}\right)\right) = \begin{pmatrix} 0 & -\mathcal{E}_{n-1}\left(u_n \tilde{\varphi}(|d_n x|)\right) \\ -\mathcal{E}_{n-1}\left(\tilde{\varphi}(|d_n x|) u_n^*\right) & 0 \end{pmatrix}.$$

The second identity follows because $\tilde{\varphi}$ is odd. Conditional symmetry (1.2) therefore gives

$$\mathcal{E}_{n-1}\left(u_n \tilde{\varphi}(|d_n x|)\right) = 0.$$

Since $\varphi(t) = \tilde{\varphi}(t)$ for every $t \ge 0$, functional calculus gives

$$\mathcal{E}_{n-1}\left(u_n \varphi(|d_n x|)\right) = 0.$$

Conversely, each $f \in C_b(\mathbb{R})$ can be decomposed as the sum of its even and odd parts, denoted by f_e and f_o. We have

$$2f_o(t) = f(t) - f(-t), \qquad \forall t \in \mathbb{R}.$$

Hence, to prove (1.2), it suffices to show

$$\widehat{\mathcal{E}}_{n-1}\left(f_o(d_n \hat{x})\right) = 0. \tag{2.5}$$

From Lemma 2.2, we have

$$\widehat{\mathcal{E}}_{n-1}\left(f_o(d_n\hat{x})\right) = \begin{pmatrix} 0 & \mathcal{E}_{n-1}\left(u_n f_o\left(|d_n(x)|\right)\right) \\ \mathcal{E}_{n-1}\left(f_o\left(|d_n(x)|\right)u_n^*\right) & 0 \end{pmatrix}.$$

By (2.3), the upper-right entry vanishes. Applying the same condition to $\overline{f_o}|_{[0,\infty)}$, we also have

$$\mathcal{E}_{n-1}\left(f_o(|d_n x|)u_n^*\right) = \mathcal{E}_{n-1}\left(u_n\overline{f_o}(|d_n x|)\right)^* = 0.$$

(iii) A direct computation gives, for every integer $k \geq 0$,

$$(d_n\hat{x})^{2k+1} = \begin{pmatrix} 0 & d_n x\left((d_n x)^* d_n x\right)^k \\ \left(d_n x\left((d_n x)^* d_n x\right)^k\right)^* & 0 \end{pmatrix}. \tag{2.6}$$

Fix n and choose $R > \|d_n x\|_\infty$. Then the spectrum of $d_n\hat{x}$ is contained in $[-R, R]$. If (1.2) holds, choose an even continuous function η of compact support which equals 1 on $[-R, R]$. Applying (1.2) to the bounded odd function $t^{2k+1}\eta(t)$ gives

$$\widehat{\mathcal{E}}_{n-1}\left((d_n\hat{x})^{2k+1}\right) = 0, \qquad k = 0, 1, 2, \ldots.$$

Hence (2.4) follows from (2.6).

Conversely, assume (2.4). Since conditional expectation preserves adjoints, (2.6) gives the vanishing of all the odd moments of $d_n\hat{x}$. For $f \in C_b(\mathbb{R})$, let f_o be its odd part. By the Stone–Weierstrass theorem, there are polynomials P_j converging uniformly to f_o on $[-R, R]$. Set

$$Q_j(t) = \frac{P_j(t) - P_j(-t)}{2}.$$

Then Q_j is odd and also converges uniformly to f_o on $[-R, R]$. Consequently,

$$\widehat{\mathcal{E}}_{n-1}\left(f_o(d_n\hat{x})\right) = \lim_{j\to\infty} \widehat{\mathcal{E}}_{n-1}\left(Q_j(d_n\hat{x})\right) = 0,$$

where the limit is in operator norm. The even part satisfies $f_e(d_n\hat{x}) = f_e(-d_n\hat{x})$, so (1.2) follows.

□

2.3. **Distribution functions and compressions.** For $x \in L_0(\mathcal{M})$, define the distribution function by

$$\lambda_t(x) = \tau\left(\chi_{(t,\infty)}(|x|)\right) \quad (t \geq 0).$$

The generalized singular-value inequalities give, for $s, t > 0$ and $x, y \in L_0(\mathcal{M})$ self-adjoint,

$$\lambda_{s+t}(x + y) \leq \lambda_s(x) + \lambda_t(y), \tag{2.7}$$

$$\lambda_t(x + y) \leq \lambda_t(x) + \lambda_0(y), \tag{2.8}$$

$$\lambda_{t+\|y\|_\infty}(x + y) \leq \lambda_t(x) \quad (y \in \mathcal{M}). \tag{2.9}$$

For these properties, see [6, Proposition 2.4 and Lemma 2.5]. For self-adjoint operators $x \leq y$, we also use the monotonicity of spectral distribution functions:

$$\tau\left(\chi_{(t,\infty)}(x)\right) \leq \tau\left(\chi_{(t,\infty)}(y)\right). \tag{2.10}$$

This is a comparison of the traces of spectral projections, not an assertion that the projections themselves are ordered. It follows by comparing the spectral subspaces above and below t, using polar decomposition; the same argument with bounded spectral windows and normality treats measurable operators. Finally, for $x \in L_p(\mathcal{M})$,

$$\|x\|_{L_p(\mathcal{M})}^p = p\int_0^\infty t^{p-1}\lambda_t(x)\,dt. \tag{2.11}$$

Lemma 2.4. *Let $\mathcal{M} \subset B(\mathcal{H})$ be a von Neumann algebra with a s.n.f. trace. Let r be a projection in $\mathcal{M}$ and let $x \in L_0(\mathcal{M})$. If $rxr = 0$, then*

$$\lambda_0(x) \le 2\tau(1-r). \tag{2.12}$$

If $h \in \mathcal{M}$ is self-adjoint, $rhr = 0$, and $\tau(1-r) < \infty$, then $h \in L_1(\mathcal{M}) \cap L_2(\mathcal{M})$ and

$$\|h\|^2_{L_2(\mathcal{M})} \le 2\tau\left((1-r)h^2\right). \tag{2.13}$$

Proof. From the definition of distribution, we have

$$\lambda_0(x) = \tau(s(|x|)) = \tau(l(x)) = \tau(r(x)). \tag{2.14}$$

Here, we use the trace property. Since $rxr = 0$, we have

$$x = (1-r)x + rx(1-r) := x_1 + x_2.$$

Obviously, $l(x_1) \le 1 - r$ and $r(x_2) \le 1 - r$. Therefore, by (2.14)

$$\begin{aligned}\lambda_0(x) &= \lambda_0(x_1 + x_2) = \tau(l(x_1 + x_2)) \\ &\le \tau(l(x_1) \vee l(x_2)) \le \lambda_0(x_1) + \lambda_0(x_2) \\ &\le 2\tau(1-r).\end{aligned}$$

This proves (2.12). Since h is bounded and its support has finite trace, h belongs to $L_1(\mathcal{M}) \cap L_2(\mathcal{M})$. By the direct sum decomposition of Hilbert space, we have

$$\mathcal{H} = r\mathcal{H} \oplus r^\perp\mathcal{H}$$

and hence,

$$h = \begin{pmatrix} rhr & rhr^\perp \\ r^\perp hr & r^\perp hr^\perp \end{pmatrix}.$$

This gives

$$\begin{aligned}\|h\|^2_{L_2(\mathcal{M})} &= 2\left\|r^\perp hr\right\|^2_{L_2(\mathcal{M})} + \left\|r^\perp hr^\perp\right\|^2_{L_2(\mathcal{M})} \\ &\le 2\left(\left\|r^\perp hr\right\|^2_{L_2(\mathcal{M})} + \left\|r^\perp hr^\perp\right\|^2_{L_2(\mathcal{M})}\right) \le 2\tau(r^\perp h^2).\end{aligned}$$

This proves (2.13). □

2.4. **A relative Golden–Thompson inequality.** In a finite von Neumann algebra the Golden–Thompson inequality states that for any self-adjoint $x, y \in \mathcal{M}$,

$$\tau(e^{x+y}) \le \tau(e^x e^y). \tag{2.15}$$

See [20, Theorem 4]; it also follows by taking $a = 0$ in [19, Theorem A]. We need a relative form which is valid also for semifinite von Neumann algebras. The next identity is inspired by [1].

Lemma 2.5. *Let $(\mathcal{M}, \tau)$ be a s.n.f. von Neumann algebra. For each $x, y \in \mathcal{M}$, we have*

$$e^x - e^y = \int_0^1 e^{(1-t)x}(x-y)e^{ty}dt. \tag{2.16}$$

Proof. Consider the operator-valued function $F(t) = e^{(1-t)x}e^{ty}$ on [0,1]. On the one hand, we have

$$F'(t) = e^{(1-t)x}(-x)e^{ty} + e^{(1-t)x}ye^{ty} = e^{(1-t)x}(y-x)e^{ty}.$$

Here, we use the commutativity of x (resp. y) and $e^{(1-t)x}$ (resp. e^{ty}). On the other hand, by the Newton–Leibniz formula, we have

$$\int_0^1 F'(t)dt = F(1) - F(0) = e^y - e^x.$$

Combining these identities completes the proof. □

Lemma 2.6. *Let $(\mathcal{M}, \tau)$ be a s.n.f. von Neumann algebra. If $x, y \in \mathcal{M} \cap L_1(\mathcal{M})$ are self-adjoint, then $e^{x+y} - 1$ and $e^x e^y - 1$ belong to $L_1(\mathcal{M})$, and*

$$\tau(e^{x+y} - 1) \le \tau(e^x e^y - 1). \tag{2.17}$$

Proof. By Hölder's inequality and (2.16), we have

$$\begin{aligned} \left\|e^x - e^y\right\|_{L_1(\mathcal{M})} &= \left\|\int_0^1 e^{(1-t)x}(x-y)e^{ty}dt\right\|_{L_1(\mathcal{M})} \\ &\le \int_0^1 e^{(1-t)\|x\|_{L_\infty(\mathcal{M})}} \|x-y\|_{L_1(\mathcal{M})} e^{t\|y\|_{L_\infty(\mathcal{M})}} dt \\ &\le e^{\|x\|_{L_\infty(\mathcal{M})} + \|y\|_{L_\infty(\mathcal{M})}} \|x-y\|_{L_1(\mathcal{M})}. \end{aligned}$$

Since $x, y \in \mathcal{M} \cap L_1(\mathcal{M})$, the above estimate gives $e^x - 1, e^y - 1, e^{x+y} - 1 \in L_1(\mathcal{M})$. Moreover,

$$e^x e^y - 1 = (e^x - 1)e^y + (e^y - 1) \in L_1(\mathcal{M}).$$

For each positive integer i, set

$$f_i = \chi_{(1/i,\infty)}(|x|), \qquad g_i = \chi_{(1/i,\infty)}(|y|), \qquad e_i = f_i \vee g_i.$$

Then $\{e_i\}_{i\ge1}$ is increasing and

$$\tau(e_i) \le \tau(f_i) + \tau(g_i) \le i\left(\|x\|_{L_1(\mathcal{M})} + \|y\|_{L_1(\mathcal{M})}\right) < \infty.$$

Put $x_i = e_i x e_i$ and $y_i = e_i y e_i$. Since $e_i \ge f_i$ and x is self-adjoint,

$$\begin{aligned} \|x - x_i\|_{L_1(\mathcal{M})} &\le \|(1-e_i)x\|_{L_1(\mathcal{M})} + \|e_i x(1-e_i)\|_{L_1(\mathcal{M})} \\ &\le 2\tau\left(|x|\chi_{(0,1/i]}(|x|)\right) \longrightarrow 0. \end{aligned}$$

Similarly, $\|y - y_i\|_{L_1(\mathcal{M})} \to 0$. Also, $\|x_i\|_{L_\infty(\mathcal{M})} \le \|x\|_{L_\infty(\mathcal{M})}$ and $\|y_i\|_{L_\infty(\mathcal{M})} \le \|y\|_{L_\infty(\mathcal{M})}$. Regard x_i and y_i as elements of the finite von Neumann algebra $\mathcal{M}_i = e_i\mathcal{M}e_i$. By (2.15),

$$\tau(e^{x_i+y_i} - e_i) \le \tau(e^{x_i}e^{y_i} - e_i),$$

where the exponentials are taken in $\mathcal{M}_i$ with respect to its identity e_i. When x_i and y_i are regarded as elements of $\mathcal{M}$, their exponentials equal 1 on $1 - e_i$. Thus the same inequality reads

$$\tau(e^{x_i+y_i} - 1) \le \tau(e^{x_i}e^{y_i} - 1).$$

The estimate at the beginning of the proof now gives $e^{x_i+y_i} - e^{x+y} \to 0$, $e^{x_i} - e^x \to 0$, and $e^{y_i} - e^y \to 0$ in $L_1(\mathcal{M})$. Furthermore,

$$\begin{aligned} \left\|e^{x_i}e^{y_i} - e^x e^y\right\|_{L_1(\mathcal{M})} &\le \|e^{x_i}\|_{L_\infty(\mathcal{M})} \left\|e^{y_i} - e^y\right\|_{L_1(\mathcal{M})} \\ &\quad + \left\|e^{x_i} - e^x\right\|_{L_1(\mathcal{M})} \|e^y\|_{L_\infty(\mathcal{M})} \longrightarrow 0. \end{aligned}$$

Taking the limit in the preceding trace inequality proves (2.17). □

3. The best constant of Burkholder-Gundy's inequality

We first prove the estimate for self-adjoint martingales satisfying (1.1). The passage to arbitrary martingales satisfying (1.2) is carried out by self-adjoint dilation at the end of the proof. The next lemma applies to ordinary self-adjoint martingale differences, without a conditional symmetry assumption.

Lemma 3.1. *Let $(\mathcal{M},\tau)$ be a s.n.f. von Neumann algebra. For each $1\le n\le N$, let $h_n\in\mathcal{M}_n\cap L_1(\mathcal{M})$ be a self-adjoint operator satisfying $\mathcal{E}_{n-1}h_n=0$. Put $H_N=\sum_{n=1}^N h_n$ and suppose that, for some $M,B\ge 0$,*

$$\|h_n\|_{L_\infty(\mathcal{M})}\le M,\qquad \sum_{n=1}^N h_n^2\le B^2\,1.$$

If $\theta>0$, $\theta M\le 1/2$, and $u=\theta t-\theta^2B^2\ge 2$, then

$$\lambda_t(H_N)\le 4\theta^2e^{-u}\|H_N\|_{L_2(\mathcal{M})}^2. \tag{3.1}$$

Proof. Observe that for each $t\ge -1/2$,

$$e^{t-t^2}\le 1+t.$$

By functional calculus, we have for each $1\le n\le N$,

$$\exp\left\{\pm\theta h_n-(\theta h_n)^2\right\}\le 1\pm\theta h_n,$$

and therefore,

$$\mathcal{E}_{n-1}\exp\left\{\pm\theta h_n-(\theta h_n)^2\right\}\le 1. \tag{3.2}$$

Put $F_0^\pm=0$ and, for each $1\le n\le N$,

$$F_n^\pm=\pm\sum_{j=1}^n\theta h_j-\sum_{j=1}^n\left(\theta h_j\right)^2.$$

Since for each $1\le n\le N$, $F_n^\pm=F_{n-1}^\pm\pm\theta h_n-(\theta h_n)^2$, Lemma 2.6 gives

$$\begin{aligned}\tau\left(\exp\{F_n^\pm\}-1\right)&\le\tau\left(\exp\left\{F_{n-1}^\pm\right\}\exp\left\{\pm\theta h_n-(\theta h_n)^2\right\}-1\right)\\&=\tau\left(\exp\left\{F_{n-1}^\pm\right\}-1\right)+\tau\left(\exp\left\{F_{n-1}^\pm\right\}\left(\exp\left\{\pm\theta h_n-(\theta h_n)^2\right\}-1\right)\right)\\&\le\tau\left(\exp\left\{F_{n-1}^\pm\right\}-1\right).\end{aligned}$$

In the last line, we apply $\mathcal{E}_{n-1}$ inside the trace and use (3.2), since $\exp\{F_{n-1}^\pm\}$ is positive and belongs to $\mathcal{M}_{n-1}$. Iterating from $F_0^\pm=0$ gives

$$\tau\left(\exp\left\{F_N^\pm\right\}-1\right)\le 0. \tag{3.3}$$

Let $\phi(t)=e^t-t-1$ for $t\in\mathbb{R}$. This function is nonnegative on $\mathbb{R}$ and increasing on $[0,\infty)$. Note that for each $1\le i<j\le N$, we have

$$\tau(h_ih_j)=\tau\left(\mathcal{E}_i(h_j)h_i\right)=0.$$

Since $\tau(h_j)=\tau(\mathcal{E}_{j-1}h_j)=0$, (3.3) and the above equality give

$$\tau\left(\phi(F_N^\pm)\right)\le\sum_{j=1}^N\tau\left(\left(\theta h_j\right)^2\mp\theta h_j\right)=\sum_{j=1}^N\theta^2\tau\left(h_j^2\right)=\theta^2\,\|H_N\|_{L_2(\mathcal{M})}^2.$$

As $\sum_{n=1}^N h_n^2\le B^2$,

$$F_N^\pm\ge\pm\theta H_N-\theta^2B^2.$$

It follows that

$$\tau\left(\chi_{(t,\infty)}(\pm H_N)\right) \le \tau\left(\chi_{(u,\infty)}(F_N^{\pm})\right) \le \frac{\theta^2}{\phi(u)} \|H_N\|_{L_2(\mathcal{M})}^2 . \tag{3.4}$$

Here, we use (2.10) and $\chi_{(u,\infty)}(t) \le \frac{\phi(t)}{\phi(u)}$. As a consequence, by $\phi(u) \ge e^u/2$, (3.4) says that

$$\lambda_t(H_N) \le \tau\left(\chi_{(t,\infty)}(H_N)\right) + \tau\left(\chi_{(t,\infty)}(-H_N)\right) \le 4\theta^2 e^{-u} \|H_N\|_{L_2(\mathcal{M})}^2 .$$

We finish this lemma. □

Let $(\mathcal{M},\tau)$ be a s.n.f. von Neumann algebra. Fix $p \ge 4$ and a self-adjoint martingale $x = \{x_n\}_{n=0}^N$ satisfying (1.1), with $x_0 = 0$. For $A > 0$, define

$$\psi_A(t) = \mathrm{sgn}(t)\min\{|t|, A\}, \qquad t \in \mathbb{R}. \tag{3.5}$$

This function is bounded, continuous, and odd. The symmetry condition therefore gives, for each $a > 0$,

$$d_n^{(a)}x = \psi_{\frac{a}{\sqrt{p}}}(d_n x), \qquad \mathcal{E}_{n-1}(d_n^{(a)}x) = 0, \qquad \|d_n^{(a)}x\|_{L_\infty(\mathcal{M})} \le \frac{a}{\sqrt{p}}. \tag{3.6}$$

The superscript (a) refers to the soft truncation at height $ap^{-1/2}$; it does not denote multiplication by a spectral projection. These differences belong to $L_p(\mathcal{M})$. We keep the square-function notation

$$S_n^{(a)}(x) = \left(\sum_{j=1}^n (d_j^{(a)}x)^2\right)^{1/2}, \qquad S_0^{(a)}(x) = 0.$$

In particular, $\left(S_n^{(a)}(x)\right)^2 \in \mathcal{M}_n \cap L_{p/2}$ and $0 \le \left(S_n^{(a)}(x)\right)^2 \le S(x)^2$.

We use the adapted version of Cuculescu's spectral recursion [5] or [17, Proposition 2.3]:

Lemma 3.2. *For each $a > 0$ and $1 \le n \le N$, let $S_n^{(a)}(x)$ be defined as above. Set*

$$q_0 = 1, \qquad q_n = q_{n-1}\,\chi_{[0,a^2]}\left(q_{n-1}\left(S_n^{(a)}(x)\right)^2 q_{n-1}\right). \tag{3.7}$$

Equivalently, the spectral projection can be taken in the reduced von Neumann algebra $q_{n-1}\mathcal{M}_n q_{n-1}$, whose identity is q_{n-1}. We have

$$q_n \in \mathcal{M}_n, \quad q_n \le q_{n-1}, \quad q_n\left(S_n^{(a)}(x)\right)^2 q_n \le a^2 q_n. \tag{3.8}$$

Moreover, q_n commutes with $q_{n-1}\left(S_n^{(a)}(x)\right)^2 q_{n-1}$.

Proof. Inductively, $q_{n-1} \in \mathcal{M}_{n-1} \subset \mathcal{M}_n$ and $q_{n-1}(S_n^{(a)}(x))^2 q_{n-1} \in \mathcal{M}_n$. Its spectral projection on $[0,a^2]$ commutes with q_{n-1}, so $q_n \in \mathcal{M}_n$ and $q_n \le q_{n-1}$. The commutation assertion and the last inequality in (3.8) follow directly from the spectral cutoff. □

Lemma 3.3. *Let $(\mathcal{M},\tau)$ be a s.n.f. von Neumann algebra. For every $a > 0$, let $\{d_n^{(a)}x\}_{n=1}^N$, $\{S_n^{(a)}(x)\}_{n=1}^N$ and $\{q_n\}_{n=0}^N$ be as above. Define*

$$g_n(a) = q_{n-1}d_n^{(a)}x\,q_{n-1}, \qquad w_N(a) = \sum_{n=1}^N g_n(a), \qquad V(a) = \sum_{n=1}^N q_{n-1}(d_n^{(a)}x)^2 q_{n-1}. \tag{3.9}$$

Then, for every $p \ge 4$

$$\sum_{n=1}^N g_n(a)^2 \le V(a) \le 2a^2\,1, \qquad \|w_N(a)\|_{L_\infty(\mathcal{M})} \le \frac{Na}{\sqrt{p}}. \tag{3.10}$$

Furthermore, $\tau(1-q_N)<\infty$ *and*

$$\tau(1-q_N)\le\frac{2}{a^2}\tau\left(S(x)^2\chi_{(a^2/2,\infty)}(S(x)^2)\right). \tag{3.11}$$

Consequently,

$$\int_0^\infty a^{p-1}\tau(1-q_N)\,da\le\frac{2^{p/2}}{p-2}\|S(x)\|_{L_p(\mathcal{M})}^p. \tag{3.12}$$

Proof. For each $1\le n\le N$, set $p_n=q_{n-1}-q_n$. (3.6) and (3.8) imply

$$\begin{aligned} q_{n-1}\left(S_n^{(a)}(x)\right)^2q_{n-1}&=q_{n-1}\left(S_{n-1}^{(a)}(x)\right)^2q_{n-1}+q_{n-1}\left(d_n^{(a)}x\right)^2q_{n-1}\\ &\le\left(1+\frac{1}{p}\right)a^2q_{n-1}\le 2a^2q_{n-1}. \end{aligned} \tag{3.13}$$

From the commutation of q_n and $q_{n-1}\left(S_n^{(a)}(x)\right)^2q_{n-1}$, we can get

$$\begin{aligned} q_{n-1}\left(d_n^{(a)}x\right)^2q_{n-1}&=p_n\left(S_n^{(a)}(x)\right)^2p_n\\ &\quad+q_n\left(S_n^{(a)}(x)\right)^2q_n-q_{n-1}\left(S_{n-1}^{(a)}(x)\right)^2q_{n-1} \end{aligned}$$

and therefore, summation gives

$$V(a)=\sum_{n=1}^N p_n\left(S_n^{(a)}(x)\right)^2p_n+q_N\left(S_N^{(a)}(x)\right)^2q_N.$$

Since $\sum_{n=1}^N p_n=1-q_N$, (3.13) yields

$$V(a)\le 2a^2(1-q_N)+a^2q_N\le 2a^2.$$

It is trivial to see that

$$g_n(a)^2=q_{n-1}d_n^{(a)}xq_{n-1}d_n^{(a)}xq_{n-1}\le q_{n-1}\left(d_n^{(a)}x\right)^2q_{n-1}$$

and

$$\|g_n(a)\|_{L_\infty(\mathcal{M})}\le\left\|d_n^{(a)}x\right\|_{L_\infty(\mathcal{M})}\le\frac{a}{\sqrt{p}}.$$

This proves (3.10).

Note that

$$\begin{aligned} \tau(p_n)&=\tau\left(q_{n-1}\chi_{(a^2,\infty)}\left(q_{n-1}\left(S_n^{(a)}(x)\right)^2q_{n-1}\right)\right)\\ &\le\frac{1}{a^p}\left\|q_{n-1}\left(S_n^{(a)}(x)\right)^2q_{n-1}\right\|_{L_{p/2}(\mathcal{M})}^{p/2}\le\frac{\|S(x)\|_{L_p(\mathcal{M})}^p}{a^p}<\infty. \end{aligned}$$

and hence, $\tau(1-q_N)<\infty$. By Hölder's inequality, we have

$$\begin{aligned} a^2\tau(1-q_N)&=\sum_{n=1}^N\tau\left(p_nq_{n-1}a^2q_{n-1}p_n\right)\le\sum_{n=1}^N\tau\left(p_n\left(S_n^{(a)}(x)\right)^2p_n\right)\\ &\le\tau\left((1-q_N)S(x)^2\right)<\infty. \end{aligned}$$

Split $S(x)^2$ as

$$S(x)^2\chi_{(0,a^2/2]}\left(S(x)^2\right)+S(x)^2\chi_{(a^2/2,\infty)}\left(S(x)^2\right).$$

Then,

$$a^2\tau(1-q_N) \le \frac{a^2}{2}\tau(1-q_N) + \tau\left(S(x)^2\chi_{(a^2/2,\infty)}\left(S(x)^2\right)\right).$$

This is (3.11). We next deal with (3.12). Let $\int_0^\infty \lambda d\mu_\lambda$ denote the spectral decomposition of $S(x)^2$. Thus, Tonelli's theorem gives

$$\begin{aligned}\int_0^\infty a^{p-1}\tau(1-q_N)da &\le \int_0^\infty\int_0^\infty 2a^{p-3}\lambda\chi_{(a^2/2,\infty)}(\lambda)d\tau(\mu_\lambda)da\\ &= \int_0^\infty 2\lambda\int_0^{\sqrt{2\lambda}} a^{p-3}da d\tau(\mu_\lambda)\\ &= \frac{2^{p/2}}{p-2}\int_0^\infty \lambda^{p/2}d\tau(\mu_\lambda) = \frac{2^{p/2}}{p-2}\|S(x)\|_{L_p(\mathcal{M})}^p.\end{aligned}$$

We complete this lemma. □

Note that the function ψ_A satisfies

$$|t-\psi_A(t)| \le \frac{t^2}{4A}, \qquad |\psi_A(t)-\psi_{A/4}(t)| \le \frac{\psi_A(t)^2}{A}. \tag{3.14}$$

For the first inequality, when $|t| \ge A$ the claim reduces to $(|t|-2A)^2 \ge 0$. For the second, the interval $A/4 \le |t| \le A$ reduces to $(|t|-A/2)^2 \ge 0$. The remaining cases follow directly from the definition.

Putting $b = a/4$, set

$$h_n(a) = q_{n-1}(a)d_n^{(b)}x\,q_{n-1}(a) - q_{n-1}(b)d_n^{(b)}x\,q_{n-1}(b), \tag{3.15}$$

$$k_n(a) = q_{n-1}(a)\left(d_n^{(a)}x - d_n^{(b)}x\right)q_{n-1}(a), \tag{3.16}$$

$$z_N(a) = \sum_n h_n(a), \qquad v_N(a) = \sum_n k_n(a). \tag{3.17}$$

Then

$$w_N(a) - w_N(b) = z_N(a) + v_N(a). \tag{3.18}$$

The projections corresponding to the two scales are not necessarily ordered. Note that both terms in $h_n(a)$ involve the same truncated martingale difference $d_n^{(b)}x$, which is essential for establishing the localized trace estimate below.

Lemma 3.4. *For every $a > 0$,*

$$\|v_N(a)\|_{L_\infty(\mathcal{M})} \le 2\sqrt{p}\,a. \tag{3.19}$$

Moreover, $\{h_n(a)\}_{n=1}^N$ are bounded self-adjoint $L_1(\mathcal{M})\cap L_2(\mathcal{M})$ martingale differences satisfying

(i)

$$\|h_n(a)\|_{L_\infty(\mathcal{M})} \le \frac{a}{2\sqrt{p}}, \quad \sum_n h_n(a)^2 \le 4a^2\,1.$$

(ii)

$$\|z_N(a)\|_{L_2(\mathcal{M})}^2 \le 8a^2\left(\tau(1-q_N(a)) + \tau(1-q_N(b))\right). \tag{3.20}$$

Whenever $u = (\sqrt{p}/a)t - 4p \ge 2$,

$$\lambda_t(z_N(a)) \le 32p\,e^{-u}\left(\tau(1-q_N(a)) + \tau(1-q_N(b))\right). \tag{3.21}$$

Proof. Recall that $d_n^{(a)}x = \psi_{\frac{a}{\sqrt{p}}}(d_n x)$. Applying the second inequality in (3.14) with $A = \frac{a}{\sqrt{p}}$, functional calculus and (3.10) give

$$\begin{aligned}\pm k_n(a) &= \pm q_{n-1}(a)\left(d_n^{(a)}x - d_n^{(b)}x\right)q_{n-1}(a)\\ &\le \frac{\sqrt{p}}{a}q_{n-1}(a)(d_n^{(a)}x)^2 q_{n-1}(a)\end{aligned}$$

and hence,

$$\pm v_N(a) \le \frac{\sqrt{p}}{a}V(a) \le 2\sqrt{p}\,a.$$

This proves (3.19).

Since $q_{n-1}(a), q_{n-1}(b) \in \mathcal{M}_{n-1}$ and $\mathcal{E}_{n-1}(d_n^{(b)}x) = 0$, the bimodule property gives $\mathcal{E}_{n-1}(h_n(a)) = 0$. Put $r = q_N(a) \wedge q_N(b)$. Then

$$rh_n(a)r = rd_n^{(b)}x\,r - rd_n^{(b)}x\,r = 0 \quad \text{and} \quad \tau(1-r) \le \tau(1-q_N(a)) + \tau(1-q_N(b)) < \infty. \tag{3.22}$$

Lemma 2.4 says that h_n belong to $L_1(\mathcal{M}) \cap L_2(\mathcal{M})$ for all $1 \le n \le N$ and (3.6) gives

$$\|h_n(a)\|_\infty \le 2\|d_n^{(b)}x\|_\infty \le a/(2\sqrt{p}).$$

Using (2.2), the compression inequality $(qDq)^2 \le qD^2q$, and $(d_n^{(b)}x)^2 \le (d_n^{(a)}x)^2$, we obtain

$$\sum_n h_n(a)^2 \le \frac{5}{4}V(a) + 5V(b) \le \frac{25}{8}a^2 \le 4a^2.$$

Orthogonality and (2.13) now imply

$$\|z_N(a)\|_2^2 = \sum_n \|h_n(a)\|_2^2 \le 2\tau\left((1-r)\sum_n h_n(a)^2\right) \le 8a^2\left(\tau(1-q_N(a)) + \tau(1-q_N(b))\right).$$

Finally, apply Lemma 3.1 with $\theta = \sqrt{p}/a$, $M = a/(2\sqrt{p})$, and $B = 2a$. Together with the above inequality, this proves (3.21). □

Lemma 3.5. *There is an absolute constant $s > 0$, independent of p, such that*

$$\int_0^\infty a^{p-1}\lambda_{s\sqrt{p}\,a}(w_N(a))\,da \le \frac{2^{p/2}}{p-2}\|S(x)\|_p^p, \qquad p \ge 4. \tag{3.23}$$

Proof. Let $L > 0$ be an absolute constant to be fixed below, and put $s = 3 + 3L/2$ and $a_j = a/4^j$. For every integer $K \ge 0$, (3.18) gives the finite identity

$$w_N(a) = \sum_{j=0}^K z_N(a_j) + \sum_{j=0}^K v_N(a_j) + w_N(a_{K+1}). \tag{3.24}$$

By (3.19),

$$\Big\|\sum_{j=0}^K v_N(a_j)\Big\|_\infty \le \sum_{j=0}^K \frac{2\sqrt{p}\,a}{4^j} \le \frac{8}{3}\sqrt{p}\,a \le 3\sqrt{p}\,a. \tag{3.25}$$

Choose

$$t_j = L\sqrt{p}\,a\,3^{-j}, \qquad t_K' = \frac{3L}{2}\sqrt{p}\,a\,3^{-(K+1)}, \qquad \sum_{j=0}^K t_j + t_K' = \frac{3L}{2}\sqrt{p}\,a.$$

By (3.10), the last term of (3.24) satisfies

$$\|w_N(a_{K+1})\|_\infty \le \frac{Na}{\sqrt{p}\,4^{K+1}} \le t'_K \quad \text{if} \quad N \le \frac{3L}{2} p(4/3)^{K+1}. \tag{3.26}$$

Choose such a K, independently of a. The distribution inequalities (2.7), (3.25) and (3.26) give

$$\begin{aligned}
\lambda_{s\sqrt{p}\,a}(w_N(a)) &= \lambda_{3\sqrt{p}\,a+\sum_{j=0}^K t_j+t'_K}\left(\sum_{j=0}^K z_N(a_j) + \sum_{j=0}^K v_N(a_j) + w_N(a_{K+1})\right) \\
&\le \lambda_{3\sqrt{p}\,a}\left(\sum_{j=0}^K v_N(a_j)\right) + \sum_{j=0}^K \lambda_{t_j}(z_N(a_j)) + \lambda_{t'_K}(w_N(a_{K+1})) \\
&= \sum_{j=0}^K \lambda_{t_j}(z_N(a_j)).
\end{aligned} \tag{3.27}$$

For each integer $j \ge 0$, replacing a by a_j in (3.21), the corresponding u_j is

$$p\left(L(4/3)^j - 4\right).$$

Taking $L \ge 5$ ensures $u_j \ge 4 > 2$. Multiplying (3.27) by a^{p-1} and integrating, (3.12) gives

$$\begin{aligned}
\int_0^\infty a^{p-1}\lambda_{s\sqrt{p}\,a}(w_N(a))\,da &\le \sum_{j=0}^\infty \int_0^\infty a^{p-1}\lambda_{t_j}(z_N(a_j))\,da \\
&\le \sum_{j=0}^\infty 32p\,e^{-u_j}\int_0^\infty a^{p-1}\left[\tau\left(1-q_N(a_j)\right)+\tau\left(1-q_N(a_j/4)\right)\right]da \\
&\le \sum_{j=0}^\infty 32p\,e^{-u_j}4^{pj}\left(4^p+1\right)\frac{2^{p/2}}{p-2}\|S(x)\|_{L_p(\mathcal{M})}^p \\
&= 32p(1+4^p)\frac{2^{p/2}}{p-2}e^{4p}\sum_{j=0}^\infty 4^{pj}e^{-Lp(4/3)^j}\|S(x)\|_{L_p(\mathcal{M})}^p.
\end{aligned} \tag{3.28}$$

Here, we use (3.21) in the second line. The third line follows from (3.12) and the change of variables

$$\int_0^\infty a^{p-1}\tau\left(1-q_N(a/4^j)\right)da = 4^{pj}\int_0^\infty a^{p-1}\tau\left(1-q_N(a)\right)da,$$

applied at j and $j+1$. Taking $L=8$, for each integer $j \ge 0$, set $b_j = 4^{pj}e^{-8p(4/3)^j}$. Then

$$\frac{b_{j+1}}{b_j} = 4^p e^{-\frac{8p}{3}(4/3)^j} \le 4^p e^{-8p/3} < \frac{1}{2}, \qquad p \ge 4, j \ge 0.$$

Consequently,

$$\sum_{j=0}^\infty b_j \le \frac{e^{-8p}}{1-4^p e^{-8p/3}} \le 2e^{-8p},$$

and hence, for $p \geq 4$

$$32p(1+4^p)e^{4p}\sum_{j=0}^{\infty} 4^{pj}e^{-8p(4/3)^j} \leq 64p(1+4^p)e^{-4p}$$
$$\leq 128p\, e^{-p(4-\log 4)} < 1,$$

where we used $p \leq e^p$, $1+4^p \leq 2\cdot 4^p$ for $p \geq 4$. This completes the proof. □

Proof of Theorem 1.1. Consider a partial sum and remainder as

$$P_N(a) = \sum_{n=1}^{N} d_n^{(a)} x, \qquad R_N(a) = x_N - P_N(a).$$

Recall that $w_N(a) = \sum_{n=1}^{N} q_{n-1} d_n^{(a)} x\, q_{n-1}$ and thus, $q_N(a)\,(P_N(a) - w_N(a))\,q_N(a) = 0$, Lemma 2.4 gives

$$\lambda_0\,(P_N(a) - w_N(a)) \leq 2\tau(1-q_N). \tag{3.29}$$

The first inequality in (3.14) gives

$$-\frac{\sqrt{p}}{4a}\,(d_n x)^2 \leq d_n x - \psi_{\frac{a}{\sqrt{p}}}(d_n x) \leq \frac{\sqrt{p}}{4a}\,(d_n x)^2$$

and hence,

$$-\frac{\sqrt{p}}{4a} S(x)^2 \leq R_N(a) \leq \frac{\sqrt{p}}{4a} S(x)^2.$$

This implies

$$\lambda_{\sqrt{p}\,a}(R_N(a)) \leq \tau\left(\chi_{(\sqrt{p}a,\infty)}\,(R_N(a))\right) + \tau\left(\chi_{(\sqrt{p}a,\infty)}\,(-R_N(a))\right) \leq 2\lambda_{4a^2}(S(x)^2). \tag{3.30}$$

Combining the distribution inequalities (2.7) and (2.8) with (3.29) and (3.30), we obtain

$$\begin{aligned}\lambda_{(s+1)\sqrt{p}\,a}(x_N) &= \lambda_{(s+1)\sqrt{p}a}\,(R_N(a) + P_N(a) - w_N(a) + w_N(a)) \\ &\leq \lambda_{s\sqrt{p}\,a}(w_N(a)) + 2\tau\,(1-q_N) + 2\lambda_{4a^2}(S(x)^2).\end{aligned} \tag{3.31}$$

Here, s is the absolute constant from Lemma 3.5. The layer-cake formula gives

$$\int_0^{\infty} a^{p-1}\lambda_{4a^2}(S(x)^2)\,da = \frac{2^{-p}}{p}\|S(x)\|_p^p.$$

Integration of (3.31), followed by (3.12) and (3.23), yields

$$\|x_N\|_p^p \leq ((s+1)\sqrt{p})^p\left[\frac{3p\,2^{p/2}}{p-2} + 2^{1-p}\right]\|S(x)\|_p^p. \tag{3.32}$$

For $p \geq 4$, the bracket is bounded by $7\cdot 2^{p/2}$, since $p/(p-2) \leq 2$. Its p-th root is uniformly bounded. The constant s is absolute, and hence (1.4) follows with an absolute C. We next prove the non-self-adjoint case.

Let $x = \{x_n\}_{n=0}^N$ satisfy (1.2). Its dilation $\hat{x} = \{\hat{x}_n\}_{n=0}^N$ is a self-adjoint conditionally symmetric L_p-martingale in $(\widehat{\mathcal{M}}, \hat{\tau})$ with $\hat{x}_0 = 0$. Recalling the computation in Lemma 2.2, we have

$$|\hat{x}_N| = \begin{pmatrix} |x_N^*| & 0 \\ 0 & |x_N| \end{pmatrix}, \qquad S(\hat{x})^2 = \begin{pmatrix} S_r(x)^2 & 0 \\ 0 & S_c(x)^2 \end{pmatrix}.$$

The equality $\tau(|x_N^*|^p) = \tau(|x_N|^p)$ follows from polar decomposition. With the normalized matrix trace,

$$\|\hat{x}_N\|_p = \|x_N\|_p, \qquad \|S(\hat{x})\|_p = \left(\frac{\|S_r(x)\|_p^p + \|S_c(x)\|_p^p}{2}\right)^{1/p}. \tag{3.33}$$

Apply the self-adjoint estimate just proved to $\hat{x}$ and then use (3.33). This proves (1.3). This proof is completed. □

Proof of Corollary 1.2. The self-adjoint martingales form a subclass of the class in Theorem 1.1, so $C_{p,\mathrm{sa}}^{\mathrm{best}} \le C_p^{\mathrm{best}}$. Conversely, apply the defining self-adjoint inequality with its best constant to $\widehat{x}$. Equation (3.33) gives

$$\|x_N\|_p \le C_{p,\mathrm{sa}}^{\mathrm{best}} \left(\frac{\|S_c(x)\|_p^p + \|S_r(x)\|_p^p}{2} \right)^{1/p} \le C_{p,\mathrm{sa}}^{\mathrm{best}} \max\{\|S_c(x)\|_p, \|S_r(x)\|_p\}.$$

Thus $C_p^{\mathrm{best}} = C_{p,\mathrm{sa}}^{\mathrm{best}}$. The same argument shows that this is also the best constant when the maximum is replaced by the averaged expression in the preceding display.

For the lower bound, take $N = \lfloor p \rfloor$ and independent Rademacher variables $r_1, \dots, r_N$. Then $x_n = \sum_{j=1}^n r_j$ is a self-adjoint conditionally symmetric martingale and $S(x) = \sqrt{N}$. The event that all signs agree has probability 2^{1-N}, so

$$\frac{\|x_N\|_p}{\|S(x)\|_p} \ge \sqrt{N}\, 2^{(1-N)/p} \ge \frac{\sqrt{N}}{2} \ge \frac{\sqrt{p}}{2\sqrt{2}}.$$

Together with the upper estimate, this proves

$$c\sqrt{p} \le C_p^{\mathrm{best}} = C_{p,\mathrm{sa}}^{\mathrm{best}} \le C\sqrt{p}, \qquad p \ge 4. \tag{3.34}$$

In particular, no bound of order $o(\sqrt{p})$ can hold uniformly over this class. This does not identify the smallest absolute coefficient in an estimate of the form $C\sqrt{p}$. □

4. Examples of noncommutative conditionally symmetric martingales

The following criterion verifies conditional symmetry for the examples. It is a sufficient condition, not an additional hypothesis in Theorem 1.1.

Proposition 4.1. *Let $x = (x_n)_{n=0}^N$ be a finite L_p-martingale with $x_0 = 0$. Suppose that, for each n, there is a normal trace-preserving $*$-automorphism α_n of $\mathcal{M}_n$ such that*

$$\alpha_n|_{\mathcal{M}_{n-1}} = \mathrm{id}, \qquad \alpha_n(d_n x) = -d_n x. \tag{4.1}$$

Then x is conditionally symmetric in the sense of Definition 2.1. In particular, when $p \ge 4$ it satisfies Theorem 1.1.

Proof. Since α_n preserves the trace and fixes $\mathcal{M}_{n-1}$, uniqueness of the trace-preserving conditional expectation gives

$$\mathcal{E}_{n-1} \circ \alpha_n = \mathcal{E}_{n-1}.$$

Put

$$\hat{\alpha}_n = \mathrm{id}_{M_2} \otimes \alpha_n, \qquad \widehat{\mathcal{E}}_{n-1} = \mathrm{id}_{M_2} \otimes \mathcal{E}_{n-1}.$$

Then

$$\widehat{\mathcal{E}}_{n-1} \circ \hat{\alpha}_n = \widehat{\mathcal{E}}_{n-1}, \qquad \hat{\alpha}_n(d_n \hat{x}) = -d_n \hat{x}.$$

Hence, for every odd $f \in C_b(\mathbb{R})$, the functional calculus and Lemma 2.2 yield

$$\widehat{\mathcal{E}}_{n-1}(f(d_n\hat{x})) = \widehat{\mathcal{E}}_{n-1}(\hat{\alpha}_n(f(d_n\hat{x}))) = \widehat{\mathcal{E}}_{n-1}(f(\hat{\alpha}_n(d_n\hat{x}))) = \widehat{\mathcal{E}}_{n-1}(f(-d_n\hat{x})).$$

Thus x is conditionally symmetric. □

4.1. **Clifford martingales.**

Example 4.2. Let (C,ρ) be the complex Clifford von Neumann algebra with canonical normalized trace and self-adjoint generators satisfying

$$c_j c_k + c_k c_j = 2\delta_{jk}1.$$

For a finite set $A = \{i_1 < \cdots < i_r\}$, put $c_A = c_{i_1}\cdots c_{i_r}$ and $c_\varnothing = 1$. The trace is characterized by $\rho(c_A) = 0$ for $A \neq \varnothing$. Let

$$C_0 = \mathbb{C}1, \qquad C_n = W^*(c_1,\dots,c_n).$$

The conditional expectation $\mathcal{E}_n$ keeps c_A when $A \subset \{1,\dots,n\}$ and annihilates it otherwise. This is the one-generator Clifford filtration considered in [15, Section 3]. Every martingale difference has the form

$$d_n x = \varphi_n(c_1,\dots,c_{n-1})c_n$$

for a polynomial φ_n. The map $\alpha_n(c_j) = c_j$ for $j < n$ and $\alpha_n(c_n) = -c_n$ preserves the Clifford relations and the canonical trace. It defines a $*$-automorphism of C_n, fixes C_{n-1}, and satisfies $\alpha_n(d_n x) = -d_n x$. Thus every Clifford martingale, whether self-adjoint or not, is conditionally symmetric by Proposition 4.1. Notice that this argument does not use a commutative formula for the powers of $\varphi_n c_n$.

Corollary 4.3. *Every finite Clifford L_p-martingale with $x_0 = 0$ and $p \geq 4$ satisfies*

$$\|x_N\|_p \leq C\sqrt{p}\,\max\{\|S_c(x)\|_p, \|S_r(x)\|_p\}.$$

For self-adjoint x, the right-hand side is $C\sqrt{p}\,\|S(x)\|_p$.

4.2. **Group von Neumann algebras.**

Example 4.4. Let $G_0 \subset G_1 \subset \cdots \subset G_N$ be a chain of discrete groups such that $[G_n : G_{n-1}] = 2$ for every $1 \leq n \leq N$. Set $\mathcal{M} = \mathrm{VN}(G_N)$ and $\mathcal{M}_n = \mathrm{VN}(G_n)$, equipped with the canonical normalized trace $\tau(a) = \langle a\delta_e, \delta_e\rangle$, where e is the group identity. Writing λ_g for the left regular unitary associated with g, the trace-preserving conditional expectation $\mathcal{E}_{n-1} : \mathcal{M}_n \to \mathcal{M}_{n-1}$ is given by

$$\mathcal{E}_{n-1}(\lambda_g) = \begin{cases} \lambda_g, & g \in G_{n-1}, \\ 0, & g \in G_n \setminus G_{n-1}. \end{cases}$$

Since G_{n-1} has index two in G_n, it is normal in G_n, and there is a character $\chi_n : G_n \to \{-1,1\}$ with $\ker\chi_n = G_{n-1}$. The map

$$\alpha_n(\lambda_g) = \chi_n(g)\lambda_g, \qquad g \in G_n,$$

defines a normal trace-preserving $*$-automorphism of $\mathcal{M}_n$ that fixes $\mathcal{M}_{n-1}$ pointwise. Indeed, it is implemented on $\ell_2(\Gamma_n)$ by the unitary $U_n\delta_g = \chi_n(g)\delta_g$, which fixes δ_e. Comparing the two maps on group unitaries yields

$$\mathcal{E}_{n-1} = \frac{1}{2}(\mathrm{id} + \alpha_n).$$

This identity extends to $L_p(\mathcal{M}_n)$, $1 \leq p < \infty$, by continuity. Consequently, every martingale difference satisfies

$$\alpha_n(d_n x) = 2\mathcal{E}_{n-1}(d_n x) - d_n x = -d_n x.$$

Proposition 4.1, applied to $x - x_0$, shows that every finite L_p-martingale with respect to this filtration is conditionally symmetric, regardless of whether it is self-adjoint. In particular, if $x_0 = 0$ and $4 \leq p < \infty$, Theorem 1.1 yields

$$\|x_N\|_p \leq C\sqrt{p}\,\max\{\|S_c(x)\|_p, \|S_r(x)\|_p\}.$$

A concrete finite-group example is provided by the dihedral group of order eight,

$$D_8 = \langle r, s : r^4 = s^2 = e,\ srs = r^{-1} \rangle,$$

with the subgroup chain

$$\{e\} \subset \langle r^2 \rangle \subset \langle r \rangle \subset D_8.$$

Each successive inclusion has index two.

For comparison, let $\mathbb{F}_\infty = \langle g_1, g_2, \ldots \rangle$ be the free group on countably many generators. Equip $\mathcal{M} = VN(\mathbb{F}_\infty)$ with its canonical trace and consider

$$\mathcal{M}_0 = \mathbb{C}1, \qquad \mathcal{M}_n = VN(\mathbb{F}(g_1, \ldots, g_n)), \qquad n \geq 1.$$

Although the successive subgroup indices are infinite, the character χ_n defined by $\chi_n(g_n) = -1$ and $\chi_n(g_j) = 1$ for $j < n$ still induces a normal trace-preserving $*$-automorphism α_n satisfying

$$\alpha_n(\lambda_{g_n}) = -\lambda_{g_n}, \qquad \alpha_n(\lambda_{g_j}) = \lambda_{g_j} \quad (j < n).$$

It fixes $\mathcal{M}_{n-1}$ pointwise. Thus the self-adjoint martingale

$$x_0 = 0, \qquad x_n = \sum_{j=1}^{n} (\lambda_{g_j} + \lambda_{g_j}^*)$$

is conditionally symmetric by Proposition 4.1. The criterion therefore applies to differences that are odd under α_n; it does not, however, establish conditional symmetry for every martingale in the free-group filtration.

4.3. **Nest filtrations.**

Example 4.5. Let $(\mathcal{M}, \tau)$ be a finite von Neumann algebra equipped with a normalized normal faithful trace. Let $\mathbb{A}$ be a totally ordered family of projections in $\mathcal{M}$ containing 0 and 1. The row nest algebra $H^r_\infty(\mathcal{M}, \mathbb{A})$ with respect to $\mathbb{A}$ is defined by

$$H^r_\infty(\mathcal{M}, \mathbb{A}) = \{x \in \mathcal{M} : xe = exe, \quad \forall e \in \mathbb{A}\}.$$

Similarly, define the column nest algebra by

$$H^c_\infty(\mathcal{M}, \mathbb{A}) = \{x \in \mathcal{M} : ex = exe, \quad \forall e \in \mathbb{A}\}.$$

Our row and column conventions are interchanged relative to those in [16]. For each $0 < p < \infty$, the noncommutative Hardy space $H^r_p(\mathbb{A})$ (resp. $H^c_p(\mathbb{A})$) is the completion of $H^r_\infty(\mathbb{A})$ (resp. $H^c_\infty(\mathbb{A})$) with respect to the $L_p(\mathcal{M})$ norm (quasi-norm when $p < 1$). In this example, we only consider the nest of order type $\mathbb{N}$, that is,

$$\mathbb{A} = \{e_k, 1\}_{k \geq 0},$$

where $\{e_k\}_{k \geq 0}$ is a sequence of increasing projections in $\mathcal{M}$ such that $e_0 = 0$ and $\vee_k e_k = 1$. Set

$$\tilde{e}_1 = e_1, \quad \tilde{e}_k = e_k - e_{k-1}, \quad k \geq 2.$$

Then $\{\tilde{e}_k\}_{k \geq 1}$ are mutually orthogonal. Let $k \geq 1$ and let

$$M_k = \left\{ x : x \in e_k \mathcal{M} e_k \quad \text{or} \quad x \in \bigcup_{n > k} \tilde{e}_n \mathcal{M} \tilde{e}_n \right\}, \quad \mathcal{M}_k = \mathrm{VN}(M_k),$$

where $\mathrm{VN}(M_k)$ denotes the von Neumann algebra generated by M_k. Then $\{\mathcal{M}_k\}_{k\geq 1}$ is a sequence of increasing von Neumann subalgebras of $\mathcal{M}$. The corresponding normal trace-preserving conditional expectation $\mathcal{E}_k : \mathcal{M} \to \mathcal{M}_k$ is given by

$$\mathcal{E}_k(x) = e_k x e_k + \sum_{n>k} \tilde{e}_n x \tilde{e}_n,$$

where the sum converges in the strong operator topology. The spaces $\{H^r_\infty(\mathcal{M}_k, \mathbb{A})\}_{k\geq 1}$ form an increasing sequence of weak-star closed, generally non-self-adjoint subalgebras of $H^r_\infty(\mathcal{M}, \mathbb{A})$; they are not in general von Neumann algebras. Since every $e \in \mathbb{A}$ belongs to $\mathcal{M}_k$, for $x \in H^r_\infty(\mathcal{M}, \mathbb{A})$ we have

$$\mathcal{E}_k(x)e = \mathcal{E}_k(xe) = \mathcal{E}_k(exe) = e\mathcal{E}_k(x)e.$$

Thus the restriction of $\mathcal{E}_k$ is a normal contractive projection onto $H^r_\infty(\mathcal{M}_k, \mathbb{A})$. The martingale filtration consists of the von Neumann algebras $\mathcal{M}_k$. For each $x \in H^r_\infty(\mathcal{M}, \mathbb{A})$ and $k \geq 2$,

$$\begin{aligned} d_k x &= \mathcal{E}_k x - \mathcal{E}_{k-1} x \\ &= e_k x e_k - e_{k-1} x e_{k-1} - \tilde{e}_k x \tilde{e}_k \\ &= \tilde{e}_k x e_{k-1} + e_{k-1} x \tilde{e}_k \\ &= \tilde{e}_k e_{k-1} x e_{k-1} + e_{k-1} x \tilde{e}_k = e_{k-1} x \tilde{e}_k. \end{aligned}$$

Here, we use the definition of $H^r_\infty(\mathcal{M}, \mathbb{A})$. Since

$$(d_k x)^* d_k x = \tilde{e}_k x^* e_{k-1} x \tilde{e}_k \in \tilde{e}_k \mathcal{M} \tilde{e}_k,$$

for every integer $l \geq 0$ we have

$$d_k x \left((d_k x)^* d_k x\right)^l \in e_{k-1} \mathcal{M} \tilde{e}_k.$$

The expectation $\mathcal{E}_{k-1}$ annihilates this off-diagonal corner. Hence

$$\mathcal{E}_{k-1}\left(d_k x \left((d_k x)^* d_k x\right)^l\right) = 0, \qquad l = 0, 1, 2, \ldots.$$

By Proposition 2.3(iii), the differences $d_k x$, $k \geq 2$, satisfy (1.2). The flip criterion also applies: the self-adjoint unitary $u_k = 1 - 2\tilde{e}_k$ belongs to $\mathcal{M}_k$, commutes with $\mathcal{M}_{k-1}$, and satisfies $u_k(d_k x)u_k = -d_k x$.

To obtain the zero initial value required in Theorem 1.1, fix $N \geq 1$ and set

$$y_0 = 0, \qquad y_n = \mathcal{E}_{n+1} x - \mathcal{E}_1 x, \qquad 1 \leq n \leq N.$$

Then y is a conditionally symmetric martingale relative to $\{\mathcal{M}_{n+1}\}_{n=0}^N$, with $d_n y = d_{n+1} x$. Consequently, for $p \geq 4$,

$$\|\mathcal{E}_{N+1} x - \mathcal{E}_1 x\|_p \leq C\sqrt{p}\max\{\|S_c(y)\|_p, \|S_r(y)\|_p\}.$$

The initial term $\mathcal{E}_1 x$ need not vanish and cannot be omitted.

4.4. **Operator-valued dyadic martingales.**

Example 4.6. Let $(\mathcal{M}, \tau)$ be a semifinite von Neumann algebra. Set

$$\Omega = \{-1, 1\}^{\mathbb{N}}, \qquad \mathbb{P} = \bigotimes_{n\geq 1} \frac{1}{2}(\delta_{-1} + \delta_1), \qquad r_n(\omega) = \omega_n,$$

and $\mathcal{F}_n = \sigma(r_1, \ldots, r_n)$, with $\mathcal{F}_0$ trivial. Define

$$\mathcal{N} = L_\infty(\Omega) \overline{\otimes} \mathcal{M}, \qquad \mathcal{N}_n = L_\infty(\mathcal{F}_n) \overline{\otimes} \mathcal{M},$$

with $\mathcal{E}_n = \mathbb{E}(\cdot \mid \mathcal{F}_n) \otimes \mathrm{id}_{\mathcal{M}}$ and trace $\varphi(f) = \int_\Omega \tau(f(\omega))\, d\mathbb{P}(\omega)$. A martingale relative to this filtration has differences

$$d_n f = r_n a_n, \qquad a_n \in L_p(\mathcal{N}_{n-1}).$$

For self-adjoint f, $a_n = a_n^*$ and

$$\mathcal{E}_{n-1}\Phi(d_n f) = \frac{\Phi(a_n) + \Phi(-a_n)}{2} = \mathcal{E}_{n-1}\Phi(-d_n f), \qquad \Phi \in C_b(\mathbb{R}).$$

Alternatively, flipping the n-th Rademacher coordinate gives the trace-preserving automorphism in Proposition 4.1, also for non-self-adjoint differences.

Corollary 4.7. *Let $p \geq 4$ and let $f = (f_n)_{n=0}^N$ be an operator-valued dyadic L_p-martingale with $f_0 = 0$. Then*

$$\|f_N\|_p \leq C\sqrt{p}\,\max\{\|S_c(f)\|_p, \|S_r(f)\|_p\}.$$

Here C is the same absolute constant as in Theorem 1.1.

References

1. N. A. Azamov, A. L. Carey, P. G. Dodds, and F. A. Sukochev, *Operator integrals, spectral shift, and spectral flow*, Canad. J. Math. **61** (2009), no. 2, 241–263. MR 2504014
2. D. L. Burkholder, *Distribution function inequalities for martingales*, Ann. Probability **1** (1973), 19–42. MR 365692
3. ———, *Sharp inequalities for martingales and stochastic integrals*, no. 157-158, 1988, Colloque Paul Lévy sur les Processus Stochastiques (Palaiseau, 1987), pp. 75–94. MR 976214
4. D. L. Burkholder and R. F. Gundy, *Extrapolation and interpolation of quasi-linear operators on martingales*, Acta Math. **124** (1970), 249–304. MR 440695
5. I. Cuculescu, *Martingales on von Neumann algebras*, J. Multivariate Anal. **1** (1971), no. 1, 17–27. MR 295398
6. T. Fack and H. Kosaki, *Generalized s-numbers of τ-measurable operators*, Pacific J. Math. **123** (1986), no. 2, 269–300. MR 840845
7. P. Hitczenko, *Comparison of moments for tangent sequences of random variables*, Probab. Theory Related Fields **78** (1988), no. 2, 223–230.
8. ———, *Upper bounds for the L_p-norms of martingales*, Probab. Theory Related Fields **86** (1990), no. 2, 225–238.
9. Y. Jiao, A. Osekowski, and L. Wu, *Noncommutative good-λ inequalities*, J. Eur. Math. Soc. (2025), Published online.
10. M. Junge and Q. Xu, *Noncommutative Burkholder/Rosenthal inequalities*, Annals of Probability **31** (2003), no. 2, 948–995.
11. ———, *On the best constants in some non-commutative martingale inequalities*, Bull. London Math. Soc. **37** (2005), no. 2, 243–253. MR 2119024
12. F. Lust-Piquard, *Inégalités de khintchine dans C_p ($1 < p < \infty$)*, C. R. Acad. Sci. Paris Sér. I Math. **303** (1986), no. 7, 289–292.
13. F. Lust-Piquard and G. Pisier, *Noncommutative Khintchine and Paley inequalities*, Arkiv för Matematik **29** (1991), no. 2, 241–260.
14. A. Osekowski, *On the best constant in the weak type inequality for the square function of a conditionally symmetric martingale*, Statistics & Probability Letters **79** (2009), 1536–1538.
15. G. Pisier and Q. Xu, *Non-commutative martingale inequalities*, Comm. Math. Phys. **189** (1997), no. 3, 667–698. MR 1482934
16. D. Potapov, F. Sukochev, and D. Zhou, *Noncommutative nest Hardy spaces and martingales*, Int. Math. Res. Not. IMRN (2024), no. 21, 13724–13757. MR 4819872
17. N. Randrianantoanina, *Non-commutative martingale transforms*, J. Funct. Anal. **194** (2002), no. 1, 181–212. MR 1929141
18. ———, *A weak type inequality for non-commutative martingales and applications*, Proceedings of the London Mathematical Society **91** (2005), no. 2, 509–544.
19. ———, *Triple operator version of the Golden-Thompson inequality for traces on von Neumann algebras*, Ann. Inst. Fourier (Grenoble) **74** (2024), no. 1, 193–233. MR 4748170
20. M. B. Ruskai, *Inequalities for traces on von Neumann algebras*, Comm. Math. Phys. **26** (1972), 280–289. MR 312284
21. G. Sadeghi and M. Moslehian, *Noncommutative martingale concentration inequalities*, Illinois J. Math. **58** (2014), no. 2.
22. Konrad Schmüdgen, *Balanced operators and operator domains*, 2021.
23. G. Wang, *Sharp square-function inequalities for conditionally symmetric martingales*, Trans. Amer. Math. Soc. **328** (1991), no. 1, 393–419.

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